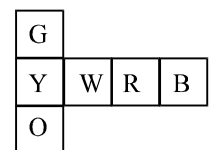


DPP - 4

- Q.1 If E & F are events with $P(E) \leq P(F)$ & $P(E \cap F) > 0$, then :
(A) occurrence of E $\Rightarrow$ occurrence of F
(B) occurrence of F $\Rightarrow$ occurrence of E
(C) non – occurrence of E $\Rightarrow$ non – occurrence of F
(D) none of the above implications holds.
- Q.2 One bag contains 3 white & 2 black balls, and another contains 2 white & 3 black balls. A ball is drawn from the second bag & placed in the first, then a ball is drawn from the first bag & placed in the second. When the pair of the operations is repeated, the probability that the first bag will contain 5 white balls is:
(A) $1/25$ (B) $1/125$ (C) $1/225$ (D) $2/15$
- Q.3 A child throws 2 fair dice. If the numbers showing are unequal, he adds them together to get his final score. On the other hand, if the numbers showing are equal, he throws 2 more dice & adds all 4 numbers showing to get his final score. The probability that his final score is 6 is:
(A) $\frac{145}{1296}$ (B) $\frac{146}{1296}$ (C) $\frac{147}{1296}$ (D) $\frac{148}{1296}$
- Q.4 A person draws a card from a pack of 52 cards, replaces it & shuffles the pack. He continues doing this till he draws a spade. The probability that he will fail exactly the first two times is :
(A) $1/64$ (B) $9/64$ (C) $36/64$ (D) $60/64$
- Q.5 Events A and C are independent. If the probabilities relating A, B and C are $P(A) = 1/5$; $P(B) = 1/6$; $P(A \cap C) = 1/20$; $P(B \cup C) = 3/8$ then
(A) events B and C are independent
(B) events B and C are mutually exclusive
(C) events B and C are neither independent nor mutually exclusive
(D) events A and C are equiprobable
- Q.6 An unbiased cubic die marked with 1, 2, 2, 3, 3, 3 is rolled 3 times. The probability of getting a total score of 4 or 6 is
(A) $\frac{16}{216}$ (B) $\frac{50}{216}$ (C) $\frac{60}{216}$ (D) none
- Q.7 A bag contains 3 R & 3 G balls and a person draws out 3 at random. He then drops 3 blue balls into the bag & again draws out 3 at random. The chance that the 3 later balls being all of different colours is
(A) 15% (B) 20% (C) 27% (D) 40%
- Q.8 A biased coin with probability P, $0 < P < 1$, of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is $2/5$, then the value of P is
(A) $1/4$ (B) $1/6$ (C) $1/3$ (D) $1/2$

- Q.9 Two numbers a and b are selected from the set of natural number then the probability that $a^2 + b^2$ is divisible by 5 is
- (A) $\frac{9}{25}$ (B) $\frac{7}{18}$ (C) $\frac{11}{36}$ (D) $\frac{17}{81}$
- Q.10 In an examination, one hundred candidates took paper in Physics and Chemistry. Twenty five candidates failed in Physics only. Twenty candidates failed in chemistry only. Fifteen failed in both Physics and Chemistry. A candidate is selected at random. The probability that he failed either in Physics or in Chemistry but not in both is
- (A) $\frac{9}{20}$ (B) $\frac{3}{5}$ (C) $\frac{2}{5}$ (D) $\frac{11}{20}$
- Q.11 When a missile is fired from a ship, the probability that it is intercepted is $\frac{1}{3}$. The probability that the missile hits the target, given that it is not intercepted is $\frac{3}{4}$. If three missiles are fired independently from the ship, the probability that all three hits the target, is
- (A) $\frac{1}{12}$ (B) $\frac{1}{8}$ (C) $\frac{3}{8}$ (D) $\frac{3}{4}$
- Q.12 An urn contains 10 balls coloured either black or red. When selecting two balls from the urn at random, the probability that a ball of each colour is selected is $\frac{8}{15}$. Assuming that the urn contains more black balls than red balls, the probability that at least one black ball is selected, when selecting two balls, is
- (A) $\frac{18}{45}$ (B) $\frac{30}{45}$ (C) $\frac{39}{45}$ (D) $\frac{41}{45}$
- Q.13 A fair die is tossed repeatedly. Mr. A wins if it is 1 or 2 on two consecutive tosses and Mr. B wins if it is 3, 4, 5 or 6 on two consecutive tosses. The probability that A wins if the die is tossed indefinitely, is
- (A) $\frac{1}{3}$ (B) $\frac{5}{21}$ (C) $\frac{1}{4}$ (D) $\frac{2}{5}$
- Q.14 An unbiased die with the numbers 1, 2, 3, 4, 6 and 8 on its six faces is rolled. After this roll if an odd number appears on the top face, all odd numbers on the die are doubled. If an even number appears on the top face, all the even numbers are halved. If the given die changes in this way then the probability that the face 2 will appear on the second roll is
- (A) $\frac{2}{18}$ (B) $\frac{3}{18}$ (C) $\frac{2}{9}$ (D) $\frac{5}{18}$
- Q.15 A butterfly randomly lands on one of the six squares of the T-shaped figure shown and then randomly moves to an adjacent square. The probability that the butterfly ends up on the R square is
- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{6}$



- Q.16 A fair coin is tossed a large number of times. Assuming the tosses are independent which one of the following statement, is True?
- (A) Once the number of flips is large enough, the number of heads will always be exactly half of the total number of tosses. For example, after 10,000 tosses one should have exactly 5,000 heads.
 - (B) The proportion of heads will be about $1/2$ and this proportion will tend to get closer to $1/2$ as the number of tosses increases
 - (C) As the number of tosses increases, any long run of heads will be balanced by a corresponding run of tails so that the overall proportion of heads is exactly $1/2$
 - (D) All of the above

[MULTIPLE OBJECTIVE TYPE]

- Q.17 Which of the following statement(s) is/are correct?
- (A) 3 coins are tossed once. Two of them atleast must land the same way. No matter whether they land heads or tails, the third coin is equally likely to land either the same way or oppositely. So, the chance that all the three coins land the same way is $1/2$.
 - (B) Let $0 < P(B) < 1$ and $P(A/B) = P(A/B^c)$ then A and B are independent.
 - (C) Suppose an urn contains 'w' white and 'b' black balls and a ball is drawn from it and is replaced along with 'd' additional balls of the same colour. Now a second ball is drawn from it. The probability that the second drawn ball is white is independent of the value of 'd'.
 - (D) A, B, C simultaneously satisfy $P(ABC) = P(A) \cdot P(B) \cdot P(C)$ and $P(AB\bar{C}) = P(A) \cdot P(B) \cdot P(\bar{C})$ and $P(A\bar{B}C) = P(A) \cdot P(\bar{B}) \cdot P(C)$ and $P(\bar{A}BC) = P(\bar{A}) \cdot P(B) \cdot P(C)$ then A, B, C are independent.
- Q.18 In each of a set of games it is 2 to 1 in favour of the winner of the previous game. What is the chance that the player who wins the first game shall win three at least, of the next four?
- Q.19 A normal coin is continued tossing unless a head is obtained for the first time. Find the probability that
- (a) number of tosses needed are at most 3.
 - (b) number of tosses are even.
- Q.20 Before a race the chance of three runners, A, B, C were estimated to be proportional to 5, 3, 2, but during the race A meets with an accident which reduces his chance to $1/3$. What are the respective chance of B and C now?
- Q.21 A is one of the 6 horses entered for a race, and is to be ridden by one of two jockeys B or C. It is 2 to 1 that B rides A, in which case all the horses are equally likely to win; if C rides A, his chance is trebled, what are the odds against his winning?